

# GIOVANNI ZHANG

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## EDUCATION

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### ETH ZURICH

#### *Master of Science in Quantum Engineering*

Zurich, CH  
2023-Present

- GPA: 5.75/6
- Selection of courses include:
  - Solid State Theory, Quantum Dot Spin Qubits, Superconducting Qubits, RF Design Techniques, Quantum Information Processing, Quantum Optics

### EPFL

#### *Bachelor of Science in Physics, Exchange semester*

Lausanne, CH  
2022-2023

- GPA: 5.75/6

### POLITECNICO DI MILANO

#### *Bachelor of Science in Physics Engineering*

Milan, IT  
2020-2023

- Graduated with 110/110 cum laude

## RESEARCH EXPERIENCE

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### QUANTUM DEVICE LAB (QUDEV), PROF. ANDREAS WALLRAFF

#### *Research Student*

Zurich, CH  
09/2025-Present

- Returned to the lab as a Research Assistant and Master's Thesis student working on 3D integrated superconducting qubits
- Designing the next generation of large scale modular devices

### ENGINEERING QUANTUM SYSTEMS GROUP, PROF. WILLIAM D. OLIVER

#### *Visiting Student (Master's Thesis)*

Boston, MA  
09/2025-09/2025

- Year-long Master's thesis project at the Massachusetts Institute of Technology (MIT) cancelled due to VISA issues
- Originally meant to work on hybrid qubit platforms and fabrication optimization of superconducting qubits.

### QUANTUM MOTION

#### *Quantum Hardware Intern*

London, UK  
02/2025-Present

- Extraction of silicon-based spin qubits' parameters at cryogenic temperatures
- Perform and improve RF reflectometry readout schemes

### QUANTUM DEVICE LAB (QUDEV), PROF. ANDREAS WALLRAFF

#### *Research Student*

Zurich, CH  
09/2024-02/2025

- Joined the research group as a student researcher to investigate Two Level Systems (TLS) coupling with superconducting qubits.
- The project is set to build the infrastructure to study E-field based AC/DC mitigation techniques of TLS in transmon qubits.
- It includes the design of the qubit parameters and on-chip Low Pass filters using in-house libraries based on GDSpy and Ansys EDT and HFSS

### QUANTUM COHERENCE LAB, PROF. DOMINIK ZUMBUHL

#### *Summer Intern*

Basel, CH  
07/2024-09/2024

- Joined the group as a summer intern working on their partnership project with Intel Corporation
- Assisted in cryogenic measurements of Intel's spin-based Tunnel Falls chip
- Worked with both dry and wet dilution refrigerators, assisting in He/N transfers and replacement of electronic components.

- Developed knowledge in QCoDes, BEOL processes (glueing, wirebonding, cleaning), and tuning quantum dots.

## ENSSLIN'S NANOPHYSICS GROUP, PROF. KLAUS ENSSLIN

Zurich, CH

### Semester Project Student

04/2024-07/2024

- Joined the group for a semester project focused on 2 parts:
  - COMSOL simulation of a Magic Angle Twisted 4-Layers Graphene – based Josephson Junction
  - Dip stick characterization of Magic Angle Twisted Bilayer Graphene devices

## AQUA LAB, PROF. EDOARDO CHARBON

Neuchatel, CH

### Research Assistant

12/2022-03/2023

- Worked in the group of Edoardo, specializing in single photon detectors
- Assisted in setting up a SPADs-based sensor for road quality assessment, embedded on FPGAs.

## AWARDS AND SCHOLARSHIPS

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- **ETH Quantum Hackathon Winner, 2024**
  - Winners of the challenge “*Quantum Neural Network for designing Differentiable Quantum Circuits to solve PDEs using PyTorch and Qadence*” proposed by PASQAL
- **“Ermenegildo Zegna Founder’s Scholarship”, 2023**
  - National scholarship for 60 recipients consisting of 15'000 CHF/year
- **“Excellence fellowship” recipient from EPFL, 2023**
  - Fellowship offered to the top 3% of applicants for a Master Program at EPFL, which I rejected in order to pursue education at ETH.
- **"Best Freshman Prize", 2021**
  - Yearly prize given to the best performing freshmen (roughly 200) of Politecnico di Milano.
- **“DSU (Diritto Scuola Universitario) Scholarship, 2020-2023**
  - Scholarship issued by the government consisting of 7'000 – 10'000 euros/year for students at Politecnico di Milano.
- **ZeroRobotics, 2018-2020**
  - National Winner (Campionato Italiano), International Finalist with EX-AEQUO 3<sup>rd</sup> place(2018), International 4th place (2019).
  - Challenge hosted by MIT and NASA on the International Space Station.
- **Math Team Olympiad, 2018-2019**
  - National Finalist (top 15), Regional Gold Medal.

## SKILLS

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- **Software Development**
  - Python/MATLAB for multipurpose tasks
  - CSS/HTML for web development
  - Software version control using Git
- **Multiphysics Simulations**
  - ANSYS package (EDT, HFSS, Q3D) for EM-field, S-parameters, and circuit elements simulations.
  - COMSOL for EM-field simulation
- **RF-circuits design**
  - QUCS for elementary circuit simulation
  - GDSpy for custom on-chip designs
- **Tiramisù Expert**
  - Both at eating and making (no complaints received so far)